

Project Design Phase-II

Data Flow Diagram & User Stories

Date	13 June 2026	Team ID	N/A (Solo Project)
Project Name	Network Anomaly Detection	Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 1 — Network Anomaly Detection System

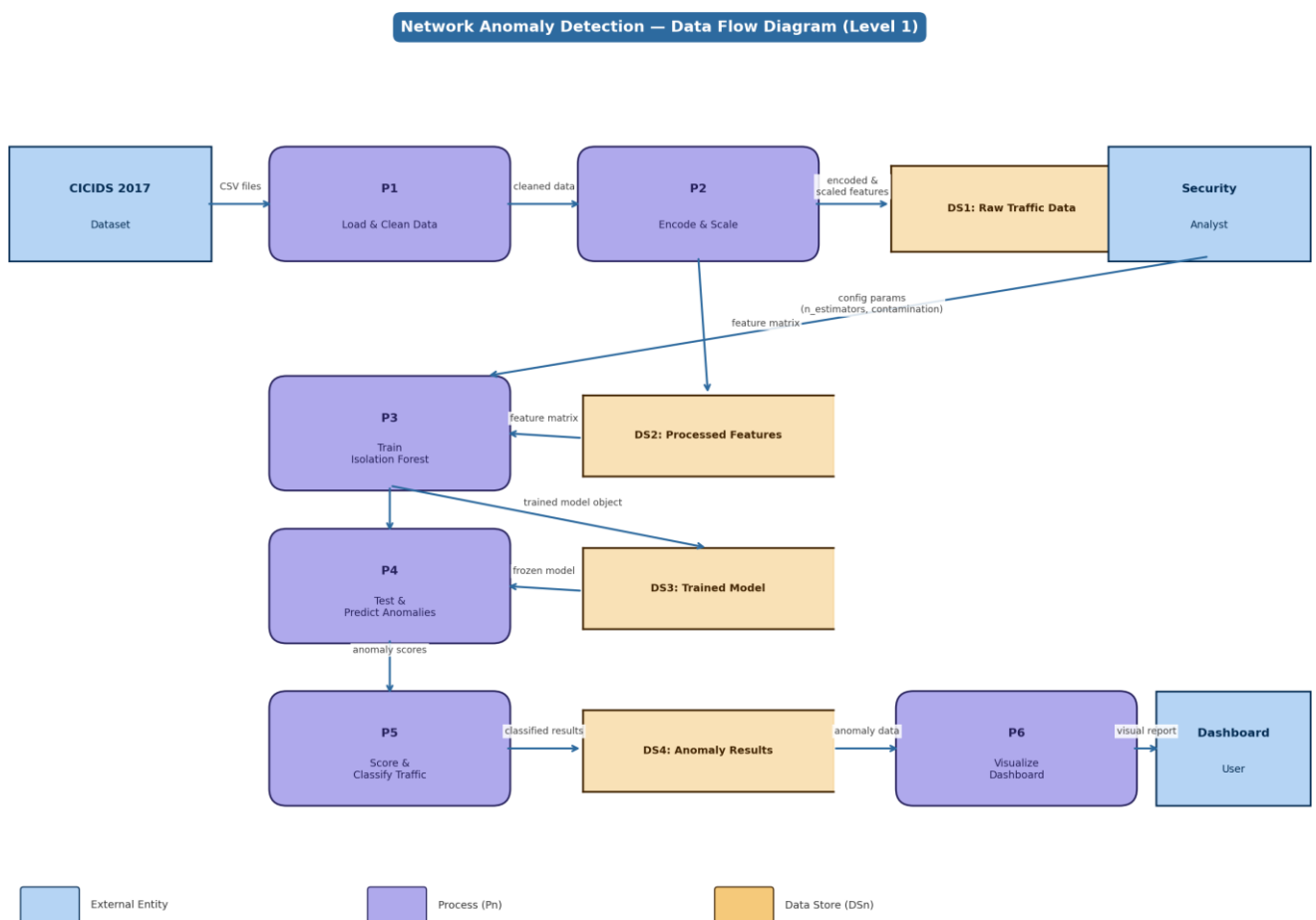


Figure 1: Level 1 DFD showing data flow from CICIDS 2017 dataset through preprocessing, model training, anomaly scoring, and visualization dashboard

DFD Elements & Descriptions

Element	ID	Name	Description
External Entity	EE1	CICIDS 2017 Dataset	Source of raw network traffic data in CSV format with 80+ features
External Entity	EE2	Security Analyst	Configures model parameters (n_estimators, contamination threshold)
External Entity	EE3	Dashboard User	End user who views the anomaly visualization dashboard
Process	P1	Load & Clean Data	Reads CSV files, removes nulls, duplicates and irrelevant columns

Process	P2	Encode & Scale Features	Converts text columns to numbers (LabelEncoder) and normalises ranges (StandardScaler)
Process	P3	Train Isolation Forest	Builds 100 isolation trees on processed training data using sklearn
Process	P4	Test & Predict Anomalies	Passes unseen traffic through the frozen trained model to get predictions
Process	P5	Score & Classify Traffic	Applies contamination threshold to anomaly scores; classifies each flow as Normal or Anomaly
Process	P6	Visualize Dashboard	Plots flagged anomalous traffic using Matplotlib/Plotly for analyst review
Data Store	DS1	Raw Traffic Data	Stores the original loaded network traffic features before processing
Data Store	DS2	Processed Features	Stores encoded and scaled feature matrix ready for model input
Data Store	DS3	Trained Model	Stores the serialised Isolation Forest model object (.pkl file)
Data Store	DS4	Anomaly Results	Stores classification output — anomaly scores and Normal/Anomaly labels per flow

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
Security Analyst	Data Ingestion	USN-1	As a security analyst, I can load the CICIDS 2017 dataset CSV files into the system.	Data loads without errors; row count matches source file	High	Sprint-1
	Data Ingestion	USN-2	As a security analyst, I can view a summary of loaded data (rows, columns, null counts).	Summary statistics are displayed correctly	Medium	Sprint-1
	Pre-processing	USN-3	As a security analyst, I can trigger data cleaning to remove nulls and duplicates.	Dataset has zero null values and no duplicate rows after cleaning	High	Sprint-1
	Pre-processing	USN-4	As a security analyst, I can encode and scale features before model training.	All features are numeric and normalised to equal ranges	High	Sprint-1
	Model Training	USN-5	As a security analyst, I can configure model parameters (n_estimators, contamination) before training.	Model trains with specified parameters without errors	High	Sprint-1
	Model Training	USN-6	As a security analyst, I can train the Isolation Forest model on the processed dataset.	Model trains successfully and is saved as a .pkl file	High	Sprint-1
	Detection	USN-7	As a security analyst, I can run anomaly detection on unseen network traffic.	Each traffic flow receives an anomaly score between 0 and 1	High	Sprint-2
	Detection	USN-8	As a security analyst, I can view which traffic flows are classified as anomalies vs normal.	Output clearly labels each flow as Normal or Anomaly	High	Sprint-2
Dashboard User	Visualization	USN-9	As a dashboard user, I can view a chart of anomaly scores across all traffic flows.	Score distribution plot renders correctly with flagged flows highlighted	High	Sprint-2
	Visualization	USN-10	As a dashboard user, I can see which features contributed most to flagged anomalies.	Top contributing features are listed or visualised per anomaly	Medium	Sprint-3
	Reporting	USN-11	As a dashboard user, I can export anomaly detection results to a CSV file.	Exported CSV contains flow ID, score, and Normal/Anomaly label	Medium	Sprint-3

User Type	Functional Requirement (Epic)	Story No.	User Story / Task	Acceptance Criteria	Priority	Release
System	Performance	USN-12	As a system, I can evaluate model performance using precision, recall, and F1-score.	Evaluation metrics are computed and displayed after each test run	High	Sprint-2